SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E9 / CAB) + GEQ

Standard-trained, fixed Val-selected best checkpoints → Full-Decodable Test (2,970 queries)

Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]

Moment-DETR · G-mIoU@1
Oracle envelope = 0.5588
Moment-DETR · G-mIoU@3
Oracle envelope = 0.5046
GMR baseline | mAP 5.47%
GMR + GEQ Full | mAP 5.47%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 12.99%
ΔmAP vs GMR: +7.52 pp (vs GEQ: +7.52 pp)
ΔG1 mean vs GMR (.4/.5/.6): +12.24 pp
ΔG3 mean vs GMR (.4/.5/.6): +10.33 pp
EaTR · G-mIoU@1
Oracle envelope = 0.5716
EaTR · G-mIoU@3
Oracle envelope = 0.5141
GMR baseline | mAP 15.19%
GMR + GEQ Full | mAP 15.26%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 15.19%
ΔmAP vs GMR: +0.00 pp (vs GEQ: -0.07 pp)
ΔG1 mean vs GMR (.4/.5/.6): +6.84 pp
ΔG3 mean vs GMR (.4/.5/.6): +7.02 pp
QD-DETR · G-mIoU@1
Oracle envelope = 0.5774
QD-DETR · G-mIoU@3
Oracle envelope = 0.5097
GMR baseline | mAP 12.65%
GMR + GEQ Full | mAP 13.22%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 14.85%
ΔmAP vs GMR: +2.20 pp (vs GEQ: +1.63 pp)
ΔG1 mean vs GMR (.4/.5/.6): +5.64 pp
ΔG3 mean vs GMR (.4/.5/.6): +5.00 pp
CG-DETR · G-mIoU@1
Oracle envelope = 0.5792
CG-DETR · G-mIoU@3
Oracle envelope = 0.5108
GMR baseline | mAP 14.67%
GMR + GEQ Full | mAP 14.85%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 16.91%
ΔmAP vs GMR: +2.24 pp (vs GEQ: +2.06 pp)
ΔG1 mean vs GMR (.4/.5/.6): +14.42 pp
ΔG3 mean vs GMR (.4/.5/.6): +14.29 pp
Sim-DETR · G-mIoU@1
Oracle envelope = 0.5839
Sim-DETR · G-mIoU@3
Oracle envelope = 0.5138
GMR baseline | mAP 19.18%
GMR + GEQ Full | mAP 20.58%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 21.05%
ΔmAP vs GMR: +1.87 pp (vs GEQ: +0.47 pp)
ΔG1 mean vs GMR (.4/.5/.6): +7.23 pp
ΔG3 mean vs GMR (.4/.5/.6): +7.10 pp
BM-DETR · G-mIoU@1
Oracle envelope = 0.5458
BM-DETR · G-mIoU@3
Oracle envelope = 0.4951
GMR baseline | mAP 3.65%
GMR + GEQ Full | mAP 4.32%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 10.91%
ΔmAP vs GMR: +7.26 pp (vs GEQ: +6.59 pp)
ΔG1 mean vs GMR (.4/.5/.6): +2.27 pp
ΔG3 mean vs GMR (.4/.5/.6): +1.49 pp
BAM-DETR · G-mIoU@1
Oracle envelope = 0.5912
BAM-DETR · G-mIoU@3
Oracle envelope = 0.5204
GMR baseline | mAP 20.62%
GMR + GEQ Full | mAP 20.26%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 21.43%
ΔmAP vs GMR: +0.81 pp (vs GEQ: +1.17 pp)
ΔG1 mean vs GMR (.4/.5/.6): +2.38 pp
ΔG3 mean vs GMR (.4/.5/.6): +2.45 pp
UVCOM · G-mIoU@1
Oracle envelope = 0.5874
UVCOM · G-mIoU@3
Oracle envelope = 0.5182
GMR baseline | mAP 6.40%
GMR + GEQ Full | mAP 8.13%
GMR + SI-Bind (E9) + GEQ (Ours) | mAP 17.37%
ΔmAP vs GMR: +10.97 pp (vs GEQ: +9.24 pp)
ΔG1 mean vs GMR (.4/.5/.6): -3.54 pp
ΔG3 mean vs GMR (.4/.5/.6): -6.34 pp
G-mIoU@1 (0–1)
G-mIoU@3 (0–1)
Existence threshold τ
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E9) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.